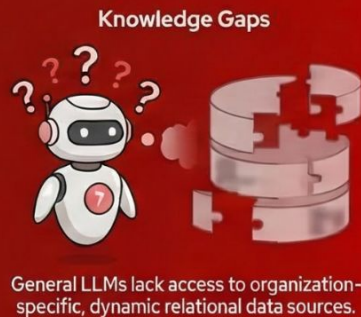
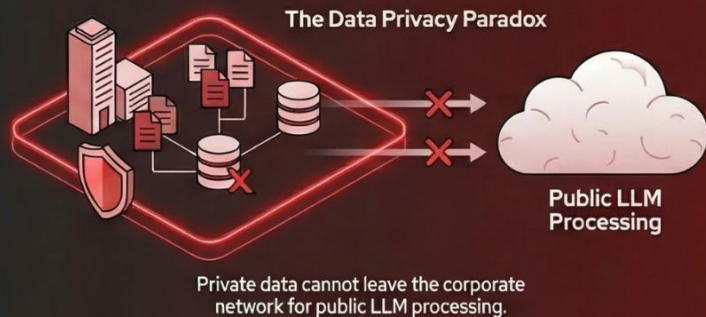


Data Governance Copilot

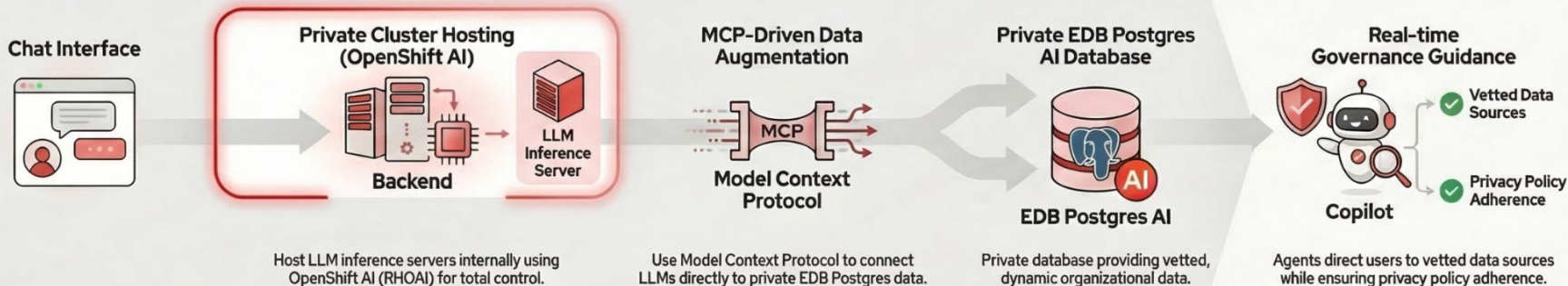
Safe conversational analytics using
EnterpriseDB's MCP Server on
Red Hat OpenShift AI

Secure AI Governance: Bridging Private Data with Enterprise LLMs

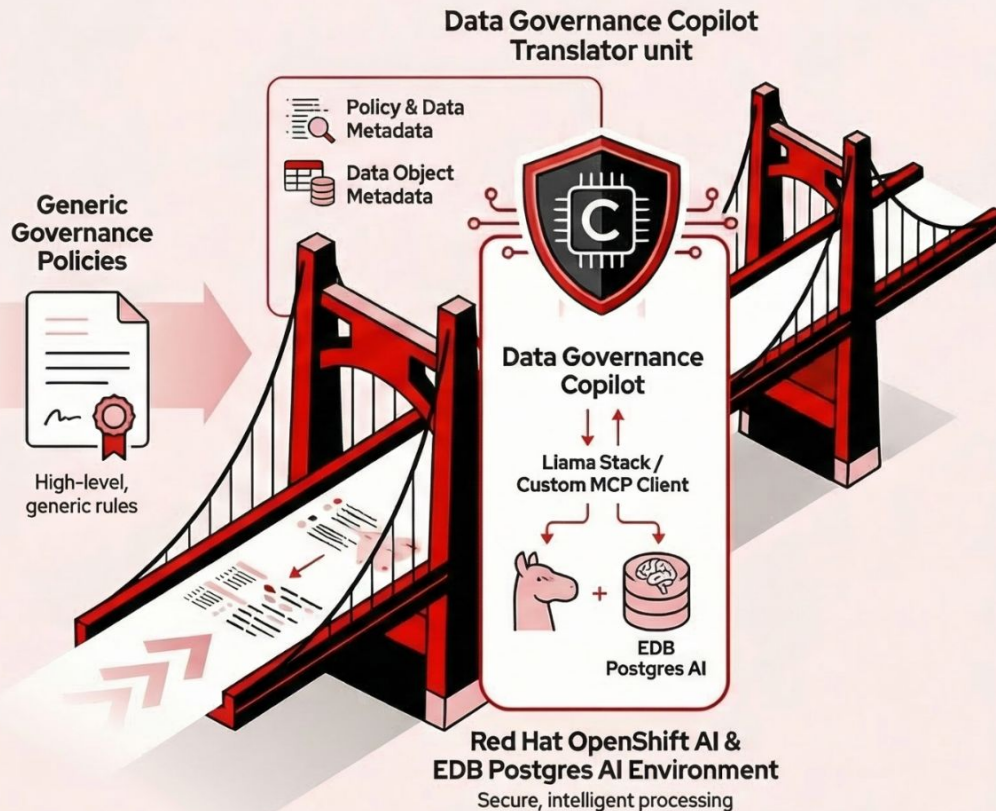
The Enterprise AI Challenge



The Secure Copilot Solution



Data Governance Copilot: Bridging the Gap Between Policy and Compliance



Intelligent User Guidance & Optimization

Directing to Vetted Sources



Proactively guides to approved, timely data

Real-time Privacy Advisory



Immediate privacy and data marking advice

Iterative Policy Fine-Tuning



Generates reasoning outputs for expert audit and adjustment

From Policy to Enforcement

Bridging Generic Policy to Metadata



Policy



Metadata

Automated PII Aggregation



PII-sensitive Table



Aggregation Only

Identifies sensitive tables, enforces aggregation



Unified Compliance Engine

Operates within OpenShift AI using EDB Postgres AI and LLMs for secure processing

Advanced AI Data Governance: Integrating EDB pg-airman-mcp with Red Hat AI Inference Server

The EDB pg-airman-mcp Toolset



execute_sql: Safe & Managed Database Access

Supports both unrestricted development access and a "Restricted Mode" that enforces read-only transactions and query time limits for production safety.

analyze_db_health: Proactive System Maintenance



Dashboard



Buffer cache



Connection Utilization



Index bloat



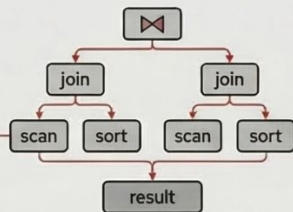
Vacuum health

Performs comprehensive checks on buffer cache hit rates, connection utilization, index bloat, and vacuum health to identify issues before they impact performance.



explain_query: Visualizing Execution Plans

Retrieves PostgreSQL execution plans and uses the hypopg extension to simulate the performance impact of hypothetical indexes without actually creating them.



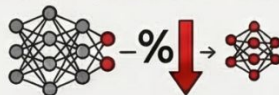
Integration Hub

EDB
Postgres AI

Data
Governance
Copilot

Optimization and Portability

Neural Magic Model Optimization



Integrates SparseGPT and quantization techniques to drastically reduce model size and memory footprint without significant accuracy loss.

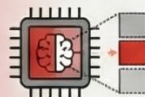
Streamlined OpenShift AI Deployment: The entire Data Governance Copilot stack, including the LLM and EDB Postgres AI database (with pgvector), is designed to run natively within OpenShift AI.

Hybrid Cloud Portability via Containerization



Delivered as a standard container image, ensuring the same inference environment runs consistently across OpenShift, RHEL, or any standard Linux system.

Red Hat AI Inference Server: The High-Performance Core



vLLM Core with PagedAttention

The serving engine uses PagedAttention to virtualize GPU memory, achieving several times higher throughput than traditional serving methods.

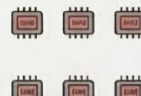
Advanced Parallelism for Massive Models



Tensor Parallelism



Pipeline Parallelism



Data Parallelism

Scales across hardware using Tensor Parallelism (intra-node), Pipeline Parallelism (inter-node), and Data Parallelism to handle massive LLM architectures.

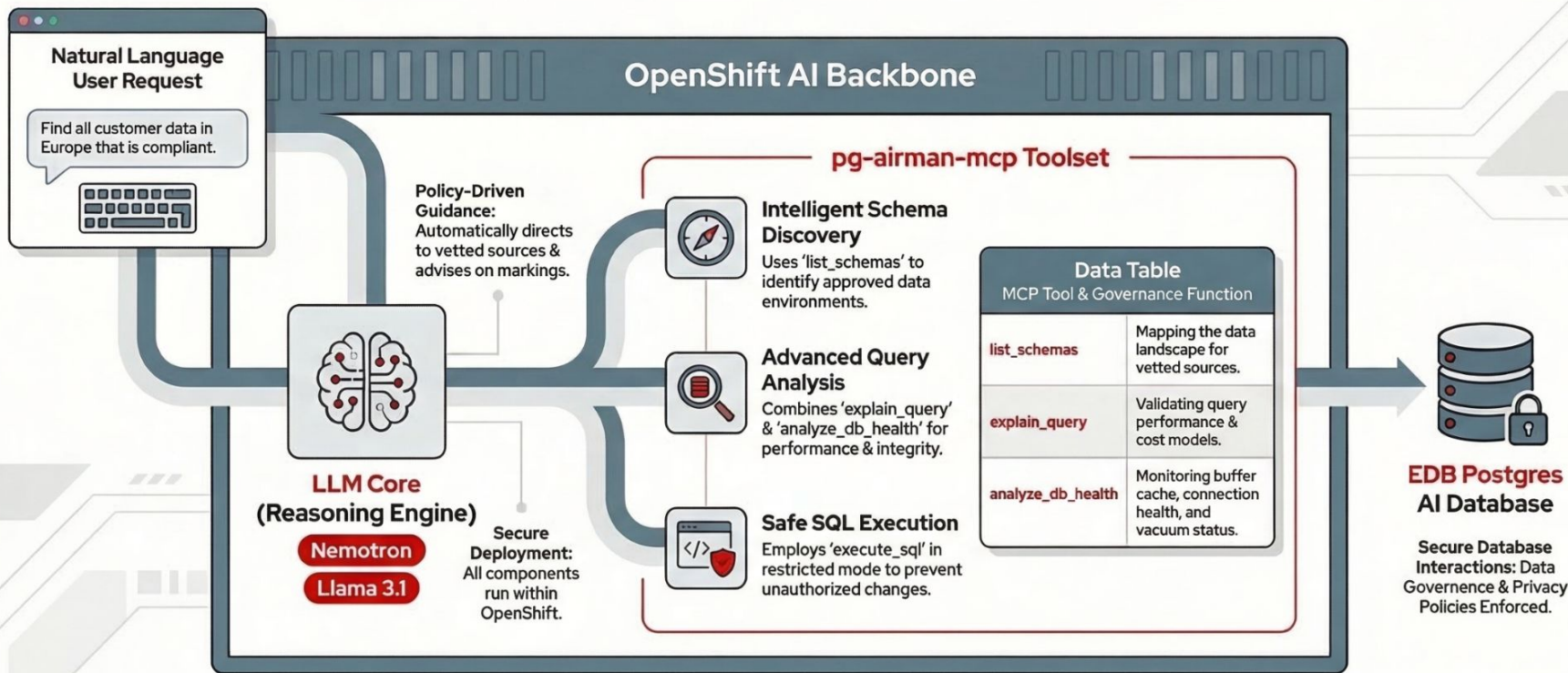


Robust Multi-Accelerator Support

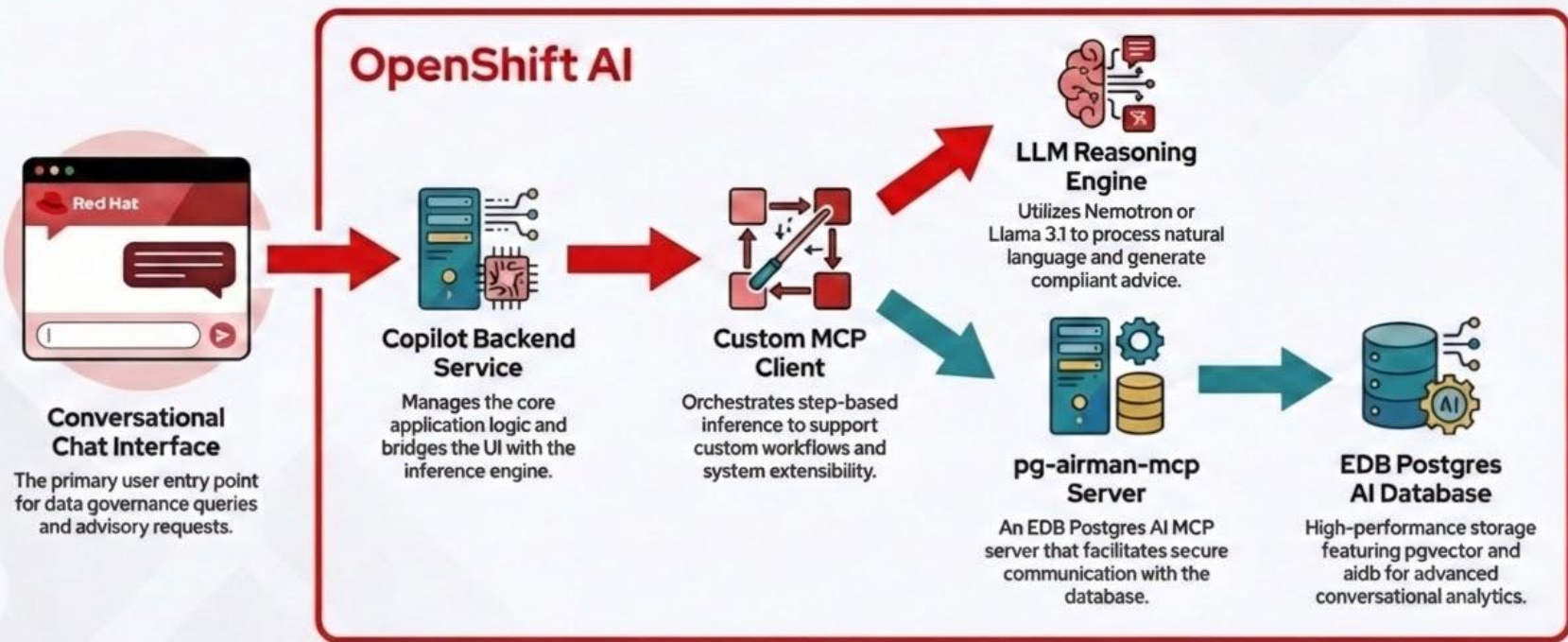
Engineered to seamlessly leverage a diverse range of hardware including NVIDIA GPUs, AMD GPUs, and Google TPUs through a unified serving layer.

Data Governance Copilot: Agentic AI for the Enterprise

Secure, Automated Data Governance and Privacy Solution via OpenShift AI, LLMs, and pg-airman-mcp

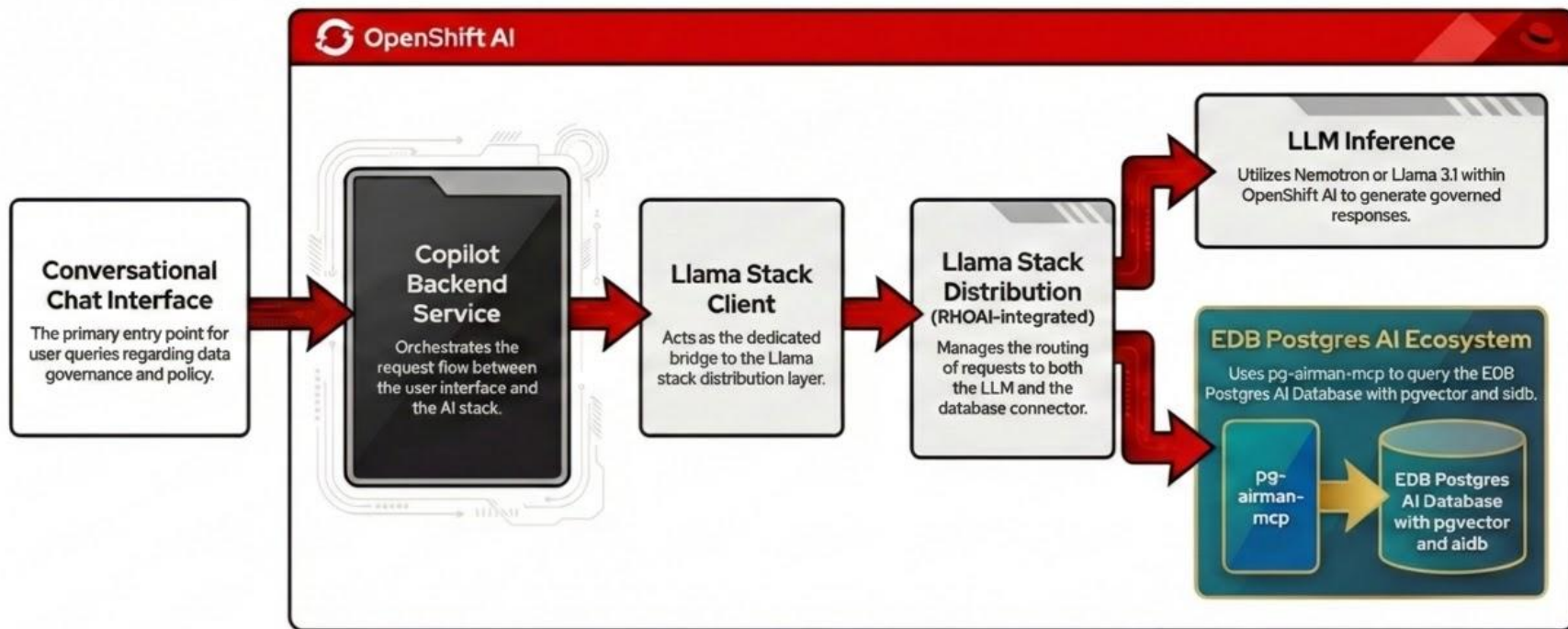


Data Governance Copilot: Custom Orchestration



Data Governance Copilot: Llama Stack Orchestration

This architecture facilitates a turn-based inference system for a conversational analytics agent. All components are hosted within OpenShift AI to provide a secure environment where users receive vetted data sources and guidance on organizational data privacy policies.



Demo Time!

